

Unit 11: Mechanical Vibrations and Oscillators

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Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

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Unit 11 at a Glance: Core Sub-Topics

This is the physics payoff of second-order theory. A mass on a spring obeys $mx'' + cx' + kx = F(t)$, and every behavior in the unit — pure oscillation, decay, blow-up, and shared rhythm — is read off that one equation by inspecting its mass, damping, stiffness, and forcing.

- **Simple harmonic motion** — Hooke's law $F = -kx$ and the natural frequency $\omega_0 = \sqrt{k/m}$.
- **Free undamped vibrations** — $x = C \cos(\omega_0 t - \phi)$ and the amplitude-phase form.
- **Damped vibrations** — the discriminant $c^2 - 4mk$ sorts the motion into over-, critically, and underdamped.
- **Forced vibrations and resonance** — driving at ω_0 produces unbounded growth; driving near it produces beats.
- **Coupled oscillators** — eigenvalues of $\mathbf{x}'' = A\mathbf{x}$ give the normal-mode frequencies.
- **The harmonic oscillator** — one equation, $x'' + \omega_0^2 x = 0$, solved four ways and hiding inside almost every stable system.

Part I — Condensed Cheat Sheet

11.1 Simple Harmonic Motion and Hooke's Law

A spring resists displacement with a force proportional to how far it is stretched and directed back toward equilibrium. That single linear law turns Newton's second law into the master oscillator

equation.

Hooke's Law and the Natural Frequency

$$F = -kx, \quad mx'' + kx = 0, \quad x'' + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{\frac{k}{m}}.$$

The period and ordinary frequency follow directly from ω_0 :

$$T = \frac{2\pi}{\omega_0}, \quad f = \frac{\omega_0}{2\pi}.$$

A hanging mass settles where gravity balances the spring, $mg = kL$, which is how a measured static stretch L delivers the stiffness $k = mg/L$.

Common Pitfall: Angular Frequency Is Not Frequency

The ω_0 inside $\cos(\omega_0 t)$ is the **angular** frequency in radians per second; the ordinary frequency f in cycles per second is smaller by a factor of 2π , $f = \omega_0/(2\pi)$. A second trap lives one step earlier: after dividing by the mass you read off ω_0^2 , *not* ω_0 . For $2x'' + 18x = 0$ the equation becomes $x'' + 9x = 0$, so $\omega_0^2 = 9$ and $\omega_0 = 3$ — never 9.

11.2 Free Undamped Vibrations

With no damping and no forcing, the imaginary roots $\pm i\omega_0$ give a clean, everlasting sinusoid. Two initial conditions — position and velocity — pin down its two constants.

General Solution and Amplitude-Phase Form

$$x(t) = c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t) = C \cos(\omega_0 t - \phi),$$

$$C = \sqrt{c_1^2 + c_2^2}, \quad \tan \phi = \frac{c_2}{c_1}.$$

The amplitude C and phase ϕ depend on the initial data, while ω_0 is fixed by the physical system alone — a stiffer spring (larger k) or a lighter mass raises ω_0 and the motion runs faster.

Solving a Spring-Mass IVP

1. Divide by the mass to reach $x'' + \omega_0^2 x = 0$ and read ω_0 .
2. Write $x = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t$.
3. Apply $x(0)$ to fix c_1 ; differentiate and apply $x'(0)$ to fix c_2 (remember $x'(0) = \omega_0 c_2$).
4. Collapse to $C \cos(\omega_0 t - \phi)$ with $C = \sqrt{c_1^2 + c_2^2}$ to read the amplitude.

11.3 Damped Vibrations and the Damping Discriminant

Add a dashpot and the equation gains a velocity term cx' . The sign of one discriminant decides the entire character of the motion.

The Damping Discriminant

$$mx'' + cx' + kx = 0, \quad \Delta = c^2 - 4mk.$$

Sign	Roots	Regime
$\Delta > 0$	two distinct real	overdamped (no oscillation)
$\Delta = 0$	repeated real	critically damped
$\Delta < 0$	complex $-b \pm i\mu$	underdamped (decaying oscillation)

The critical value $c = 2\sqrt{mk}$ is the boundary; it returns the system to equilibrium fastest *without* overshooting.

Common Pitfall: The Quasi-Frequency Is Not ω_0

An underdamped solution $e^{-bt}(c_1 \cos \mu t + c_2 \sin \mu t)$ oscillates at the **quasi-frequency** $\mu = \sqrt{\omega_0^2 - b^2}$, which is strictly *below* the undamped natural frequency ω_0 . Damping always slows the oscillation; it never speeds it up.

11.4 Forced Vibrations, Beats, and Resonance

Drive the system with $F_0 \cos(\omega t)$ and the steady response depends sharply on how the drive frequency ω compares to the natural frequency ω_0 .

Steady Amplitude, Beats, and Resonance

For undamped forced motion $mx'' + kx = F_0 \cos(\omega t)$ with $\omega \neq \omega_0$,

$$x_p(t) = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t).$$

As $\omega \rightarrow \omega_0$ the denominator vanishes and the amplitude blows up. At exact **resonance** $\omega = \omega_0$ the trial gains a factor of t ,

$$x_p(t) = t(A \cos \omega_0 t + B \sin \omega_0 t),$$

so the amplitude grows linearly without bound. For ω *close to but not equal* to ω_0 the two nearby frequencies interfere to produce **beats** — a fast oscillation under a slow envelope whose period lengthens as $\omega \rightarrow \omega_0$.

Common Pitfall: Resonance Versus Beats

Beats and resonance are not the same phenomenon. **Beats** need $\omega \approx \omega_0$ (close, not equal) and give a *bounded*, slowly modulated amplitude. **Resonance** needs $\omega = \omega_0$ exactly and gives *unbounded* growth from the resonant t factor. Mistaking one for the other misreads whether a structure merely throbs or is being torn apart.

11.5 Coupled Oscillators and Normal Modes

When two masses share a spring, neither equation stands alone: each acceleration depends on the other's position. Writing the system as $\mathbf{x}'' = \mathbf{A}\mathbf{x}$ turns the physics into an eigenvalue problem.

Normal Modes from Eigenvalues

$$\mathbf{x}'' = A\mathbf{x}, \quad A\mathbf{v} = \lambda\mathbf{v}, \quad \omega^2 = -\lambda, \quad \omega = \sqrt{-\lambda}.$$

Each eigenvalue gives one **normal mode**: a motion in which all masses oscillate at a single shared frequency with a fixed amplitude ratio (the eigenvector). A system with n degrees of freedom has exactly n modes, and the general motion is a **superposition** of them, each with its own amplitude and phase.

Finding the Normal Modes

1. Assemble the coupling matrix A from the equations of motion.
2. Solve $\det(A - \lambda I) = 0$ for the eigenvalues λ .
3. Convert each to a mode frequency $\omega = \sqrt{-\lambda}$.
4. Find each eigenvector for the mode shape (amplitude ratio), then superpose.

11.6 The Harmonic Oscillator Solved Four Ways

Strip away damping and forcing and one equation remains, $x'' + \omega_0^2 x = 0$. It is worth meeting from several directions because it is the linear heartbeat of nearly every stable system.

One Equation, Many Faces

$$x'' + \omega_0^2 x = 0 \implies r^2 + \omega_0^2 = 0 \implies r = \pm i\omega_0 \implies x = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t.$$

The energy method — multiply by x' and integrate — yields the conserved quantity

$$\frac{1}{2}(x')^2 + \frac{1}{2}\omega_0^2 x^2 = \text{const.}$$

The same equation governs an ideal LC circuit $Lq'' + \frac{1}{C}q = 0$ with $\omega_0 = 1/\sqrt{LC}$, because near any stable equilibrium a smooth potential looks quadratic and its restoring force looks linear.

Unit 11 Key Takeaway

Everything flows from $m\mathbf{x}'' + c\mathbf{x}' + k\mathbf{x} = F(t)$. The **natural frequency** $\omega_0 = \sqrt{k/m}$ sets the rhythm; the **discriminant** $c^2 - 4mk$ sorts damped motion into over-, critically, and under-damped; **resonance at** $\omega = \omega_0$ forces a t factor and unbounded growth while a near-match gives **beats**; and the **eigenvalues of** $\mathbf{x}'' = A\mathbf{x}$ unlock the normal modes of coupled motion. Underneath it all sits the **harmonic oscillator** $x'' + \omega_0^2 x = 0$, the universal linear model of a system near stable equilibrium.

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb through simple harmonic motion setups, free undamped initial-value problems, the amplitude-phase collapse, Hooke's law calibration, the three damping regimes sorted by the discriminant, the critical damping coefficient, resonance identification, and a coupled-oscillator capstone. Solutions follow in Part III.

1. **(SHM setup)** For $2x'' + 18x = 0$, find the natural angular frequency ω_0 , the period T , and the ordinary frequency f .
2. **(Free undamped IVP)** Solve $x'' + 9x = 0$ with $x(0) = 0$, $x'(0) = 6$.
3. **(Amplitude-phase form)** Write $x(t) = 5 \cos 2t + 12 \sin 2t$ as a single sinusoid $C \cos(2t - \phi)$; give the amplitude C and phase ϕ .
4. **(Hooke's law calibration)** A weight of $mg = 10$ N stretches a spring by $L = 0.2$ m. Find the stiffness k , and with $g = 10$ m/s² find the resulting natural frequency ω_0 .
5. **(Damping — overdamped)** Solve $x'' + 5x' + 6x = 0$, evaluate the discriminant, and name the damping regime.
6. **(Damping — critically damped)** Solve $x'' + 2x' + x = 0$ and identify the regime from the discriminant.
7. **(Damping — underdamped)** Solve $x'' + 2x' + 5x = 0$, name the regime, and state the quasi-frequency.
8. **(Critical damping coefficient)** For $m = 1$ and $k = 9$, find the damping coefficient c that makes the system critically damped.
9. **(Resonance identification)** For $x'' + 25x = \cos(\omega t)$, find the forcing frequency ω that produces resonance, and compute the resonant particular solution at that frequency.
10. **(Coupled oscillators, capstone)** For $x_1'' = -2x_1 + x_2$ and $x_2'' = x_1 - 2x_2$, find the normal-mode frequencies from the eigenvalues $\lambda = -1, -3$, identify the mode shapes, and write the general solution.

Part III — Complete Solutions

Solution 1. Divide by the mass 2 to reach standard form:

$$2x'' + 18x = 0 \implies x'' + 9x = 0 \implies \omega_0^2 = 9, \omega_0 = 3.$$

Then the period and ordinary frequency follow from ω_0 :

$$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{3}, \quad f = \frac{\omega_0}{2\pi} = \frac{3}{2\pi}.$$

Reading ω_0 off the *undivided* coefficient 18 is the classic error; the mass divides out first.

Solution 2. The roots of $r^2 + 9 = 0$ are $\pm 3i$, so $x = c_1 \cos 3t + c_2 \sin 3t$. Apply the data:

$$x(0) = c_1 = 0, \quad x'(t) = 3c_2 \cos 3t, \quad x'(0) = 3c_2 = 6 \Rightarrow c_2 = 2.$$

$$x(t) = 2 \sin 3t.$$

The derivative pulls down a factor of $\omega_0 = 3$, so c_2 is $6/3 = 2$, not 6.

Solution 3. Combine the coefficients into one amplitude:

$$C = \sqrt{c_1^2 + c_2^2} = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$$

The phase satisfies $\tan \phi = c_2/c_1 = 12/5$ with both components positive (first quadrant):

$$\phi = \arctan \frac{12}{5} \approx 1.176 \text{ rad}, \quad x(t) = 13 \cos(2t - 1.176).$$

Solution 4. The static balance $mg = kL$ gives the stiffness directly:

$$k = \frac{mg}{L} = \frac{10}{0.2} = 50 \text{ N/m}.$$

With $g = 10 \text{ m/s}^2$ the weight $mg = 10 \text{ N}$ gives mass $m = 1 \text{ kg}$, so

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{50}{1}} = 5\sqrt{2} \approx 7.07 \text{ rad/s}.$$

Solution 5. The characteristic equation factors:

$$r^2 + 5r + 6 = (r + 2)(r + 3) = 0 \Rightarrow r = -2, -3.$$

The discriminant $\Delta = c^2 - 4mk = 25 - 24 = 1 > 0$ confirms two distinct real roots, so the motion is **overdamped** — pure decay, no oscillation:

$$x(t) = c_1 e^{-2t} + c_2 e^{-3t}.$$

Solution 6. The characteristic equation is a perfect square:

$$r^2 + 2r + 1 = (r + 1)^2 = 0 \Rightarrow r = -1 \text{ (double)}.$$

Here $\Delta = 4 - 4 = 0$, the boundary case, so the system is **critically damped** and the repeated root carries a factor of t :

$$x(t) = (c_1 + c_2 t) e^{-t}.$$

Solution 7. The quadratic formula on $r^2 + 2r + 5 = 0$ gives complex roots:

$$r = \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i.$$

With $\Delta = 4 - 20 = -16 < 0$ the motion is **underdamped**: a decaying envelope times an oscillation,

$$x(t) = e^{-t}(c_1 \cos 2t + c_2 \sin 2t).$$

The quasi-frequency is $\mu = 2$, just below the undamped value $\omega_0 = \sqrt{5} \approx 2.236$.

Solution 8. Critical damping is the boundary $\Delta = 0$, i.e. $c^2 = 4mk$, so

$$c = 2\sqrt{mk} = 2\sqrt{(1)(9)} = 2 \cdot 3 = 6.$$

Any $c > 6$ overdamps the system and any $c < 6$ underdamps it.

Solution 9. From $x'' + 25x = 0$ the natural frequency is $\omega_0 = \sqrt{25} = 5$, so resonance occurs at

$$\omega = \omega_0 = 5.$$

At resonance the bare trial $A \cos 5t + B \sin 5t$ already solves the homogeneous equation, so it gains a factor of t : try $x_p = t(A \cos 5t + B \sin 5t)$. Differentiating twice,

$$x_p'' = -25t(A \cos 5t + B \sin 5t) + 10(-A \sin 5t + B \cos 5t),$$

so $x_p'' + 25x_p = 10(-A \sin 5t + B \cos 5t)$. Matching to $\cos 5t$:

$$10B = 1, \quad -10A = 0 \Rightarrow A = 0, \quad B = \frac{1}{10}, \quad x_p(t) = \frac{t}{10} \sin 5t.$$

The t out front is the signature of resonance: the amplitude grows without bound.

Solution 10 (Capstone). In matrix form $\mathbf{x}'' = A\mathbf{x}$ with

$$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}, \quad \lambda = -1, -3.$$

Each mode frequency comes from $\omega^2 = -\lambda$:

$$\omega_1 = \sqrt{-(-1)} = 1, \quad \omega_2 = \sqrt{-(-3)} = \sqrt{3}.$$

The eigenvectors give the mode shapes. For $\lambda = -1$, $(A + I)\mathbf{v} = 0$ yields $\mathbf{v}_1 = (1, 1)^\top$, the **in-phase** mode (lower frequency); for $\lambda = -3$, $(A + 3I)\mathbf{v} = 0$ yields $\mathbf{v}_2 = (1, -1)^\top$, the **out-of-phase** mode (higher frequency, since it stretches the coupling spring more). The general motion is a superposition of the two modes:

$$\mathbf{x}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} (a_1 \cos t + b_1 \sin t) + \begin{pmatrix} 1 \\ -1 \end{pmatrix} (a_2 \cos \sqrt{3}t + b_2 \sin \sqrt{3}t).$$

Unit 11 Key Takeaway

Read the motion straight off the coefficients. The **natural frequency** $\omega_0 = \sqrt{k/m}$ and the collapse to $C \cos(\omega_0 t - \phi)$ govern free motion; the **discriminant** $c^2 - 4mk$ sorts damping into **overdamped** (> 0), **critically damped** ($= 0$, $c = 2\sqrt{mk}$), and **underdamped** (< 0); **resonance** at $\omega = \omega_0$ injects the t factor that drives $x_p = \frac{t}{10} \sin 5t$ to grow without bound; and the **eigenvalues** of $\mathbf{x}'' = A\mathbf{x}$ deliver the normal-mode frequencies $\omega = \sqrt{-\lambda}$ whose superposition is the full coupled motion.